

3 ATM Interface :

```
import java.util.ArrayList;
import java.util.List;
import java.util.Scanner;

class BankAccount {
    private double balance;
    private List<String> transactionHistory;

    public BankAccount(double initialBalance) {
        balance = initialBalance;
        transactionHistory = new ArrayList<>();
        transactionHistory.add(String.format("Account opened with balance: Rs.
%.2f", balance));
    }

    public double getBalance() {
        return balance;
    }

    public void deposit(double amount) {
        if (amount <= 0) {
            System.out.println("Deposit amount must be positive.");
            return;
        }
    }
}
```

```
    balance += amount;
    transactionHistory.add(String.format("Deposited: Rs. %.2f", amount));
}
```

```
public boolean withdraw(double amount) {
    if (amount <= 0) {
        System.out.println("Withdrawal amount must be positive.");
        return false;
    }
    if (amount <= balance) {
        balance -= amount;
        transactionHistory.add(String.format("Withdrew: Rs. %.2f", amount));
        return true;
    }
    return false;
}
```

```
public void printTransactionHistory() {
    System.out.println("Transaction History:");
    for (String transaction : transactionHistory) {
        System.out.println(transaction);
    }
}
```

```
public boolean transferTo(BankAccount targetAccount, double amount) {
    if (amount <= 0) {
```

```

        System.out.println("Transfer amount must be positive.");
        return false;
    }
    if (amount <= balance) {
        balance -= amount;
        targetAccount.deposit(amount);
        transactionHistory.add(String.format("Transferred: Rs. %.2f to another
account", amount));
        return true;
    }
    return false;
}
}

```

```

class ATM {
    private BankAccount account;
    private Scanner scanner;

    public ATM(BankAccount bankAccount) {
        account = bankAccount;
        scanner = new Scanner(System.in);
    }

```

```

    public void displayMenu() {
        System.out.println("\nATM Menu:");
        System.out.println("1. Check Balance");
        System.out.println("2. Deposit");
    }

```

```
System.out.println("3. Withdraw");
System.out.println("4. Transfer Funds");
System.out.println("5. Transaction History");
System.out.println("6. Exit");
}
```

```
private double getPositiveAmount(String prompt) {
    double amount = -1;
    while (amount <= 0) {
        System.out.print(prompt);
        if (scanner.hasNextDouble()) {
            amount = scanner.nextDouble();
            if (amount <= 0) {
                System.out.println("Amount must be positive. Please try again.");
            }
        } else {
            System.out.println("Invalid input. Please enter a numeric value.");
            scanner.next(); // consume invalid input
        }
    }
    return amount;
}
```

```
public void run() {
    while (true) {
        displayMenu();
    }
}
```

```
System.out.print("Select an option: ");

if (!scanner.hasNextInt()) {

    System.out.println("Invalid input. Please enter a number between 1
and 6.");

    scanner.next(); // consume invalid input
    continue;
}

int choice = scanner.nextInt();


switch (choice) {

    case 1:

        System.out.printf("Your balance: Rs. %.2f%n",
account.getBalance());

        break;

    case 2:

        double depositAmount = getPositiveAmount("Enter amount to
deposit: Rs. ");

        account.deposit(depositAmount);

        System.out.printf("Deposit successful. Your balance: Rs. %.2f%n",
account.getBalance());

        break;

    case 3:

        double withdrawAmount = getPositiveAmount("Enter amount to
withdraw: Rs. ");

        if (account.withdraw(withdrawAmount)) {

            System.out.printf("Withdrawal successful. Your balance: Rs.
%.2f%n", account.getBalance());

        } else {
```

```

        System.out.println("Insufficient balance.");
    }
    break;
case 4:
    double transferAmount = getPositiveAmount("Enter amount to
transfer: Rs. ");
    // For demonstration, create a dummy target account
    BankAccount targetAccount = new BankAccount(0);
    if (account.transferTo(targetAccount, transferAmount)) {
        System.out.printf("Transfer successful. Your balance: Rs. %.2f%n",
account.getBalance());
        System.out.printf("Recipient's balance: Rs. %.2f%n",
targetAccount.getBalance());
    } else {
        System.out.println("Insufficient balance for transfer.");
    }
    break;
case 5:
    account.printTransactionHistory();
    break;
case 6:
    System.out.println("Thank you for using the ATM!");
    scanner.close();
    return;
default:
    System.out.println("Invalid option. Please select a valid option.");
}

```

```
    }  
  }  
}
```

```
public class Main {  
    public static void main(String[] args) {  
        BankAccount userAccount = new BankAccount(0); // Initial balance set to 0  
        ATM atm = new ATM(userAccount);  
        atm.run();  
    }  
}
```